EXP.NO:21

Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

PROBLEM STATEMENT:

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

CODE

```c
#include <stdio.h>
#define MAX 10000
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    int seen[MAX]={0};
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++){
        seen[a[i]]++;
    }
    for(int i=0;i<n;i++){
        if(seen[a[i]]>1){
            printf("%d",a[i]);
            return 0;
        }
    }
}
```

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OUTPUT:



	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.